

Your first day is already a crisis.

Solace is growing fast — 150 people, a wellness app adding subscribers every week. Nobody agrees on how many. You were hired this morning to replace the last Analytics Engineer, who left three weeks ago without a handover. Your inbox already has 40 unread messages. Most of them are **numbers that don't match.**

UNLOCKS: THE AUDIT INSTINCT

1

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MEET THE CAST

Four people who are about to need you

M

Maya
CEO

"I just need ONE number I can say out loud in the board meeting."

P

Priya
Head of Growth

"Our acquisition dashboard says we're up 40% this month. Somebody check this?"

D

Diego
Head of Product

"Retention looks fine on my end. Not sure why Finance is panicking."

F

Farrah
CFO

"None of these dashboards match my spreadsheet. I don't trust any of them."

You'll hear from all four of them by the end of this level.

INVESTIGATION

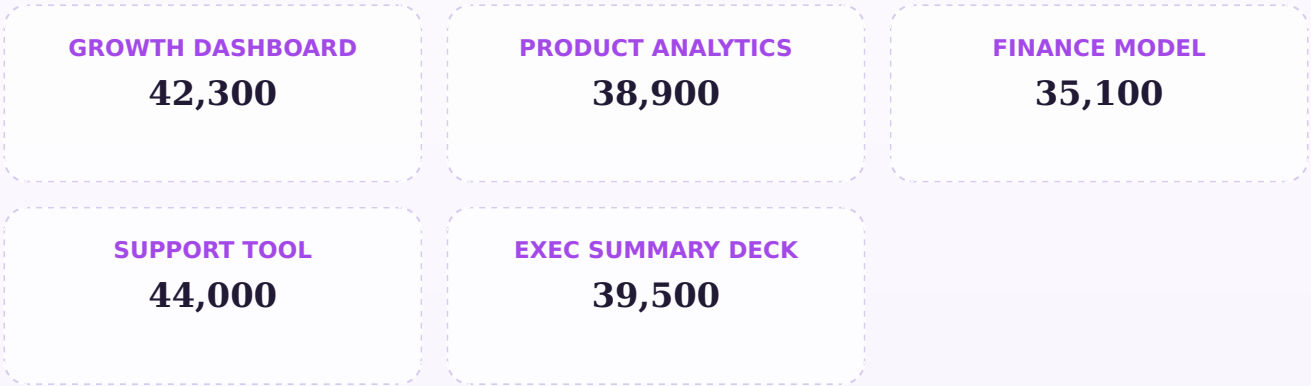
The mess is never random

Solace isn't broken because everyone is bad at their job. It's broken because five departments built five separate spreadsheets, each with its own definition of "active user," "signup," and "revenue" — and nobody ever agreed on which one is the truth. This happens at almost every fast-growing company. Your job isn't to find who's lying. Nobody is. Your job is to find where the definitions quietly diverged.

- 1. Ask for the definition, not the number.** "What counts as active?" matters more than the number itself.
- 2. Find the source, not the summary.** A dashboard is a claim. The underlying table is the evidence.
- 3. Assume good faith, expect divergence.** Every team optimized their own spreadsheet for their own question — that's exactly why they don't match.

Mapping the spreadsheet sprawl

Before you touch a query, map what already exists. Five sources, five numbers, same month, same company:



Five numbers. Same month. Same company. This is where Level 1 begins.

Three departments, three different "active users"

CONFIDENTIAL

Growth says **42,300** active users. Product says **38,900**. Finance says **35,100**. All three pulled their number this morning, for the same month.

Your prediction — why don't these match?

Write your guess in before you turn the page — the answer's waiting on the other side, not right below.



CASE CRACKED

Why the numbers don't match

Recap: Growth 42,300 · Product 38,900 · Finance 35,100.

Growth counts anyone who opened the app once. Product counts anyone who completed a full session. Finance counts anyone who was still a paying subscriber at month's end. All three are "correct" by their own definition — none of them is wrong, and none of them is "the" active user number. That's the actual bug: nobody wrote the definition down anywhere everyone could see it.

REFLECTION CARDS

Before you move on

What would you do differently?

What assumption did Growth, Product, and Finance each make without realizing it?

What could go wrong if you just picked one number and moved on?

From someone who's seen this before

I've seen a company report two different revenue numbers to its own board, in the same meeting. Nobody noticed for six months. This mistake is invisible until it's expensive.

I made this exact mistake early on — I picked the number that made my dashboard look best, not the one that was most defensible. A CFO caught it live, in a meeting. I never made that mistake again.

Real interview question, if you're prepping:
"Tell me about a time two stakeholders disagreed on a metric. What did you do?" You just lived the answer.

BOSS FIGHT

The All-Hands Meeting

Maya has called an all-hands. She wants to know, in front of Priya, Diego, and Farrah, which "active users" number is right. All eyes are on you — the new hire, on day one.

☐ **A — Pick Finance's number publicly, since it's the most conservative.**

Trade-off: fast, but you haven't actually solved anything — the next report will disagree again.

☐ **B — Say "there isn't one right answer yet" and propose a working session to define it together.**

Trade-off: slower, feels weaker in the room, but it's the only option that actually fixes the root cause.

☐ **C — Ask for 24 hours to investigate before answering.**

Trade-off: buys you credibility-building time, but risks looking evasive if you don't come back with something concrete.

There's no wrong pick here — but B is the one that actually starts fixing Solace's data, not just this meeting. Real Analytics Engineers get judged on whether the same argument happens again next month.

UNLOCK

Your first collectible

Before You Trust Any Number

1. Ask for the definition, not the number.
2. Find the source, not the summary.
3. Assume good faith, expect divergence.

The Curiosity Framework Cheat Card — keep this one, you'll use it in every level after this.

Rookie mistakes & pro secrets

ROOKIE MISTAKE

Reporting a number in a meeting because it's the one you were handed, without asking who calculated it or how.

PRO ANALYST SECRET

The most trusted Analytics Engineers are the ones who say "let me confirm the definition" out loud, in the room, even when it's uncomfortable. It looks like uncertainty. It's actually the opposite.

Level 1, condensed

The framework: ask for the definition, find the source, assume good faith and expect divergence.

Solace's 3 numbers: 42,300 (opened app once) · 38,900 (completed a session) · 35,100 (paying at month's end).

The Boss Fight call: propose a working session instead of picking a winner.

Still unresolved: nobody's written the definitions down anywhere — that's Level 4's problem to solve for good.

Test Yourself on Solace's Data

No skipping ahead, no peeking at the next page early. Pick your answers below, then turn the page to see how you did.

1. Which team's number counts someone as active the moment they open the app once?

- ☐ A. Finance ☐ B. Growth ☐ C. Product

2. If Solace wanted the most conservative (smallest) active-user count, whose definition should it use?

- ☐ A. Growth ☐ B. Support ☐ C. Finance

3. What was actually broken at Solace?

- ☐ A. A wrong number ☐ B. A missing definition ☐ C. A broken database

Your verdict — and the answer key — is one page away.

CASE FILE VERIFIED

How'd you do?

1 → B, Growth. Growth counts anyone who opened the app once — the loosest definition.

2 → C, Finance. 35,100 was the lowest number because "still paying at month's end" is the tightest bar.

3 → B, a missing definition. Every team's math was correct by its own rule — nobody had agreed on which rule counted.

Got all 3?

Case closed. You're ready for Level 2: The SQL Forest.

Got 2 out of 3?

Sharp instincts — go reread the one that surprised you before moving on.

Got 1 or fewer?

No shame in it. Flip back to the Detective Case and the Boss Fight, then try again — this one's worth actually nailing.

Apply It to Your Own World

This part has no answer key — the whole point is that it's your metric, not Solace's. Open your own case file below.

Your Case

Metric	Where it comes from	How it's defined	What could make it wrong
CLUE #1			
CLUE #2			
CLUE #3			

This one goes on your Career Checklist — it's the first real artifact of your Level 8 portfolio project.

LEVEL COMPLETE



World 1: The Spreadsheet Kingdom

You finished Level 1 and survived your first day at Solace.

Maya doesn't know you well yet, but Farrah just added you to the Finance Slack channel. That's not nothing.

"Every argument about a number is really an argument about a definition nobody wrote down."

NEXT UP

Level 2: The SQL Forest

You've mapped the mess. Now you get database access — and the ability to check any number yourself.